

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 3.1: Unsupervised Learning

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Das Interdisziplinäre Digitale
Labor der Universität Graz



Motivation

Supervised Learning

- Known relationship between attributes and target values (classes or numbers).
- Prediction of target values for new observations.

How can we model data when no target values are known?

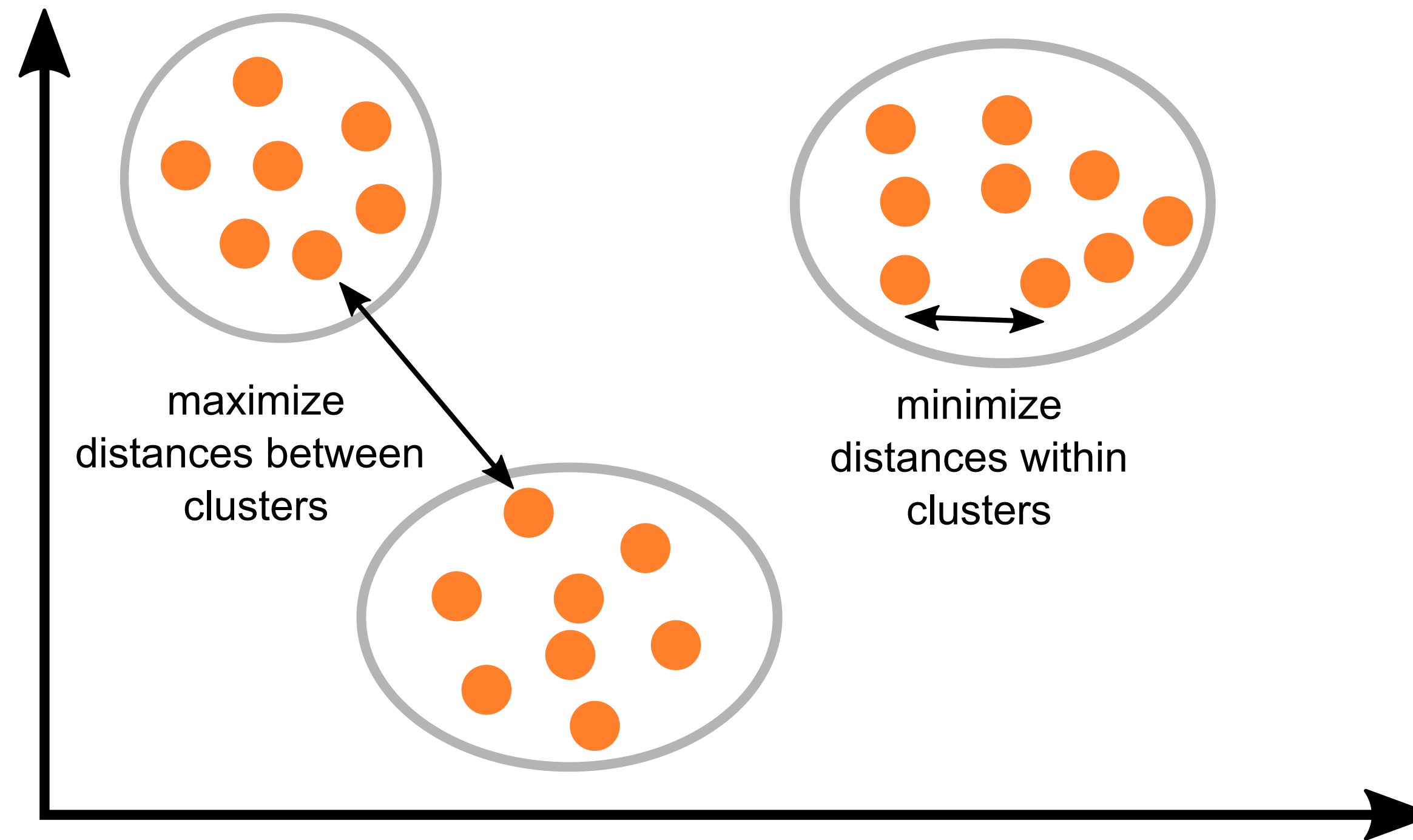
→ Unsupervised Learning (cluster analysis)

Parents' Annual Income Attribute	School Grade Target Value
80,000 €	
180,000 €	
50,000 €	
160,000 €	
60,000 €	
170,000 €	
60,000 €	

Example: Predicting School Success.

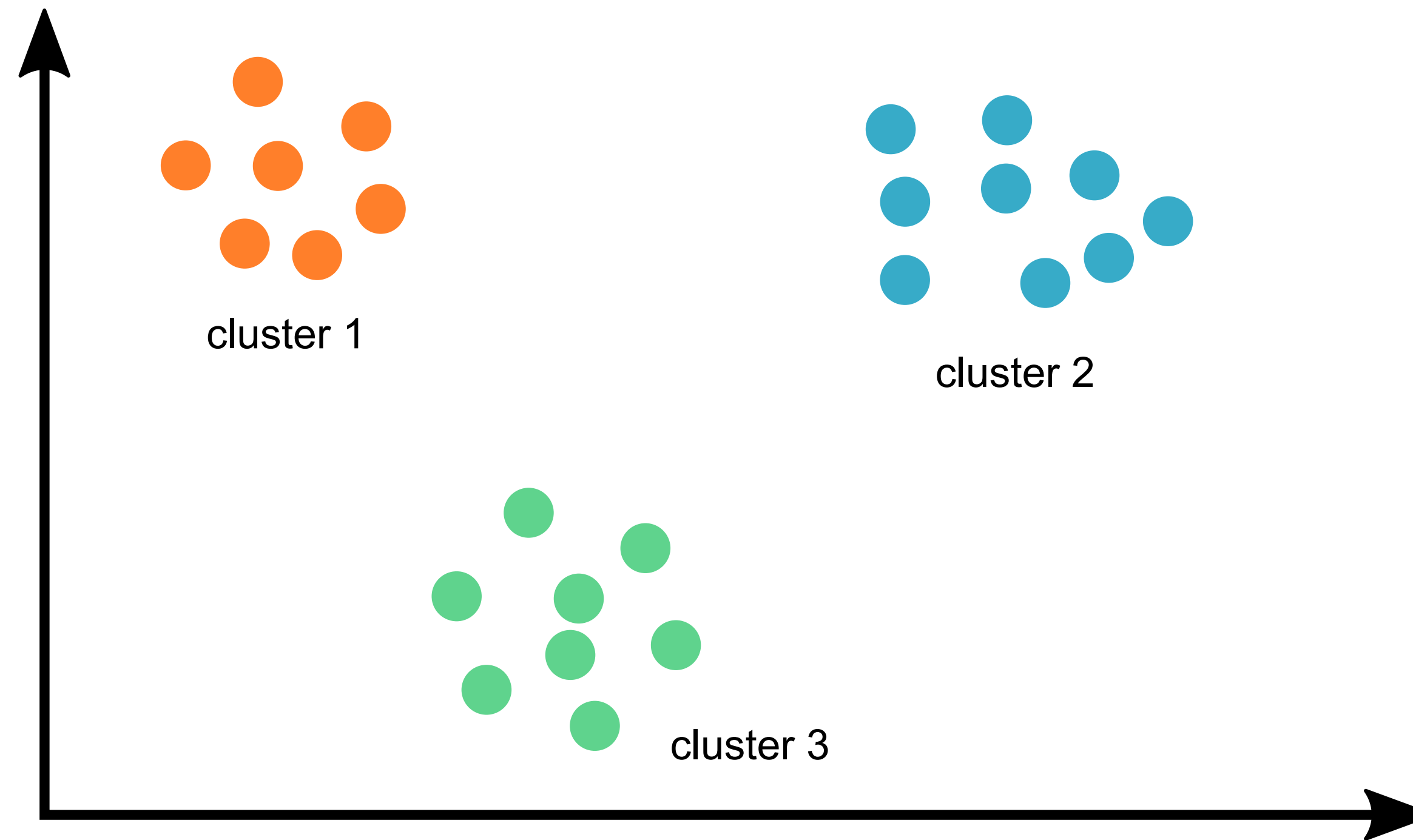
What is Cluster Analysis?

Task: Group similar points.



What is Cluster Analysis?

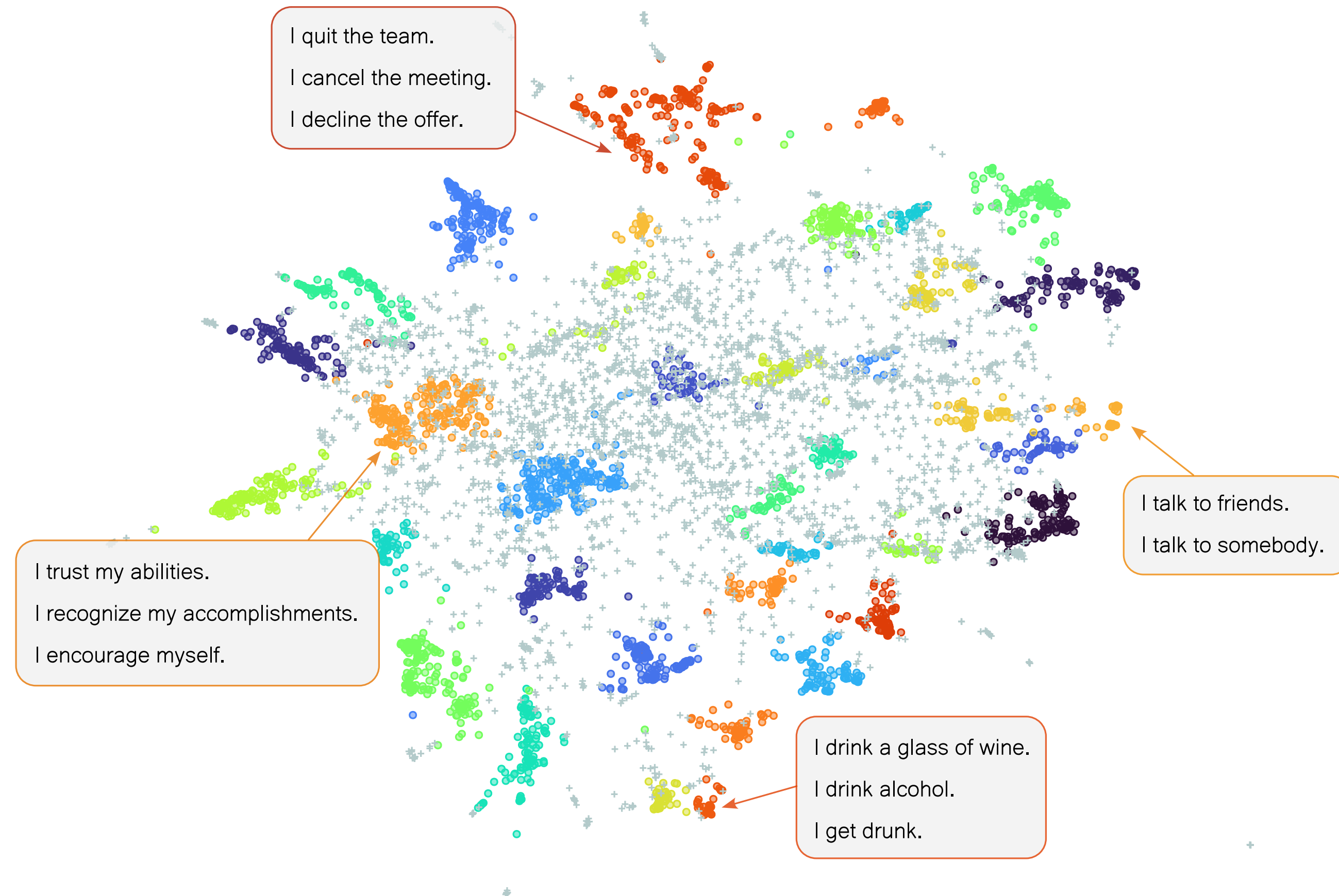
Task: Group similar points.



A **cluster** is a set of data points.
A **clustering** is a set of clusters.

Applications of Cluster Analysis: Finding Patterns

Question: What strategies do people have for emotion regulation?



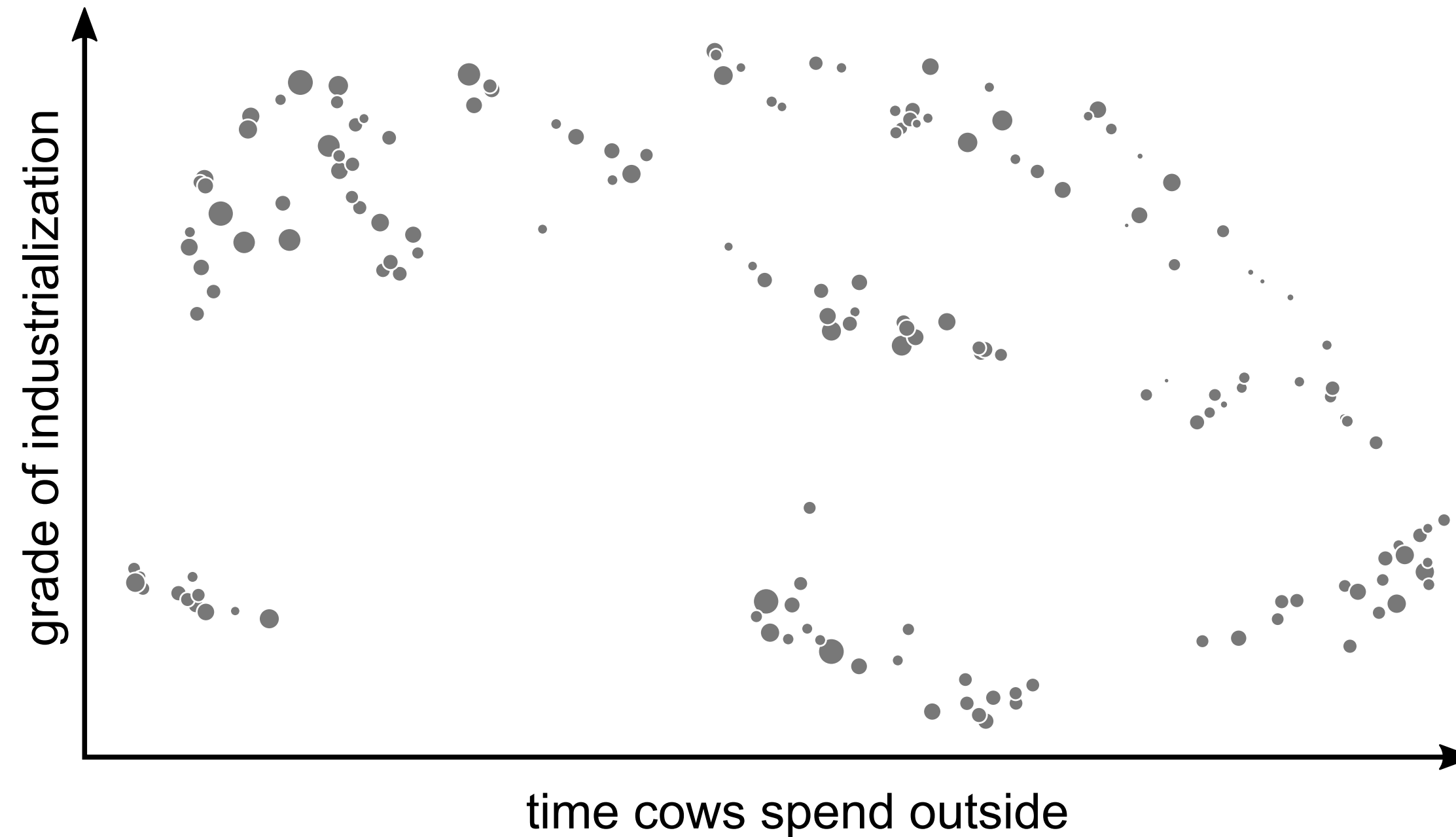
Note

- To display texts as in the figure, we first need to convert them into numbers.
- This is done using embeddings → more on this in Lecture 4!

From: Herderich et al., [A Computational Method to Reveal Psychological Constructs from Text Data](#), *Psychol. Methods* (2024).

Applications of Cluster Analysis: Summarizing Data

Dairy farms in Austria:
Which farms have a high disease risk?



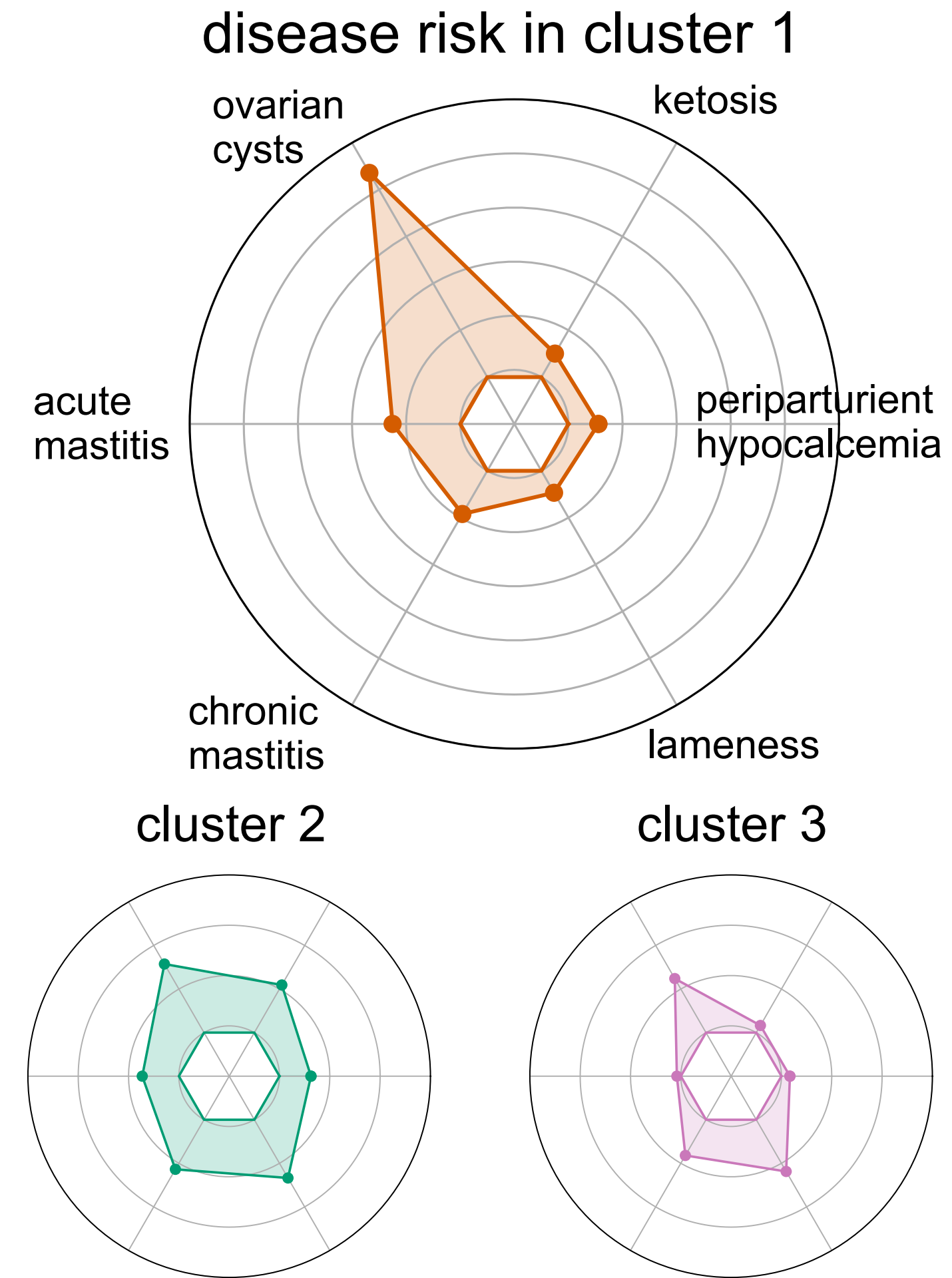
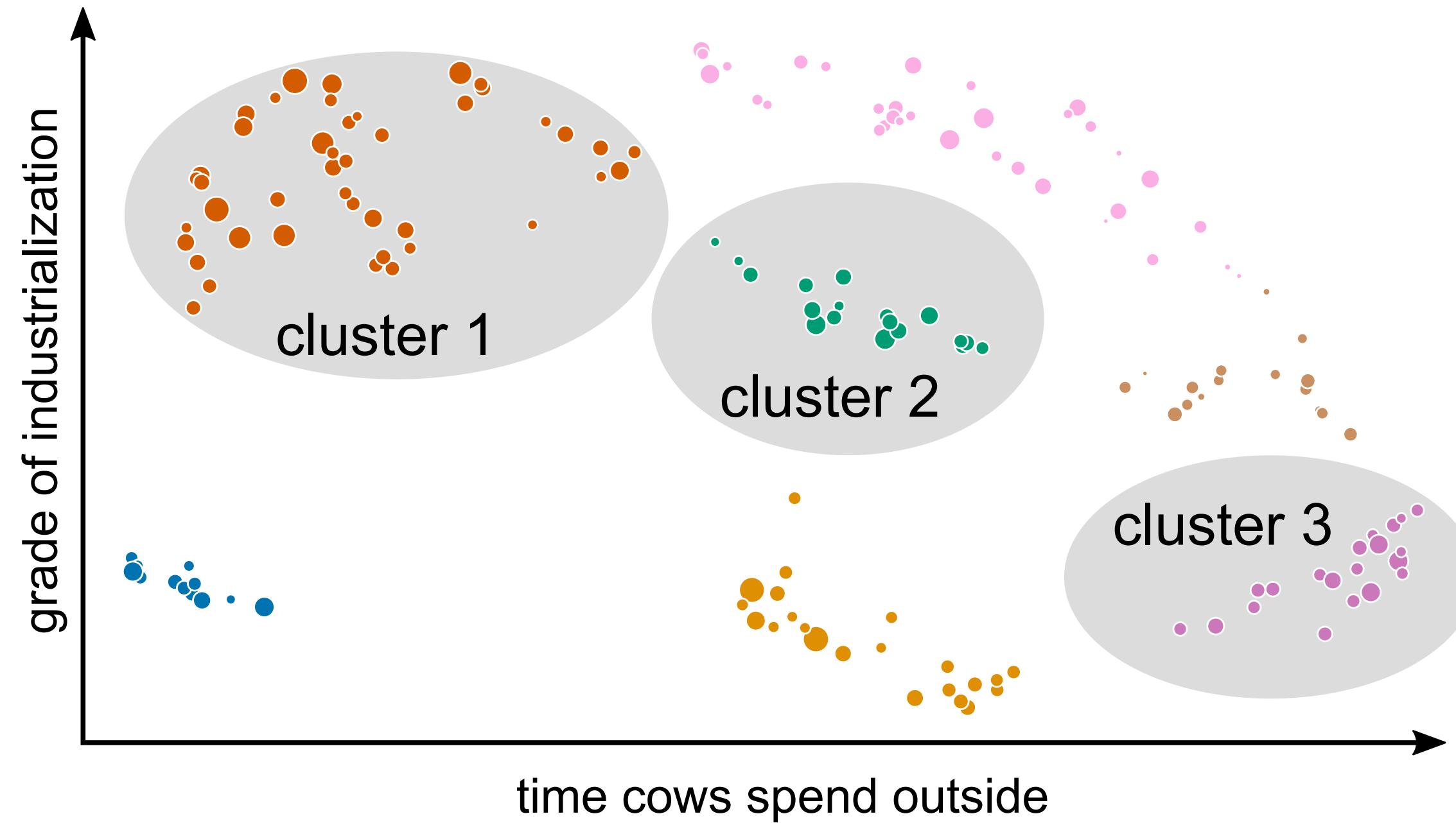
Note

- In order to represent **high-dimensional** data, i.e. data with many attributes in two dimensions in a meaningful way, we must first apply **dimension reduction**.
- We will cover this later in today's lecture.

From: Matzhold et al., [A systematic approach to analyse the impact of farm-profiles on bovine health](#), *Scientific Reports* (2021).

Applications of Cluster Analysis: Summarizing Data

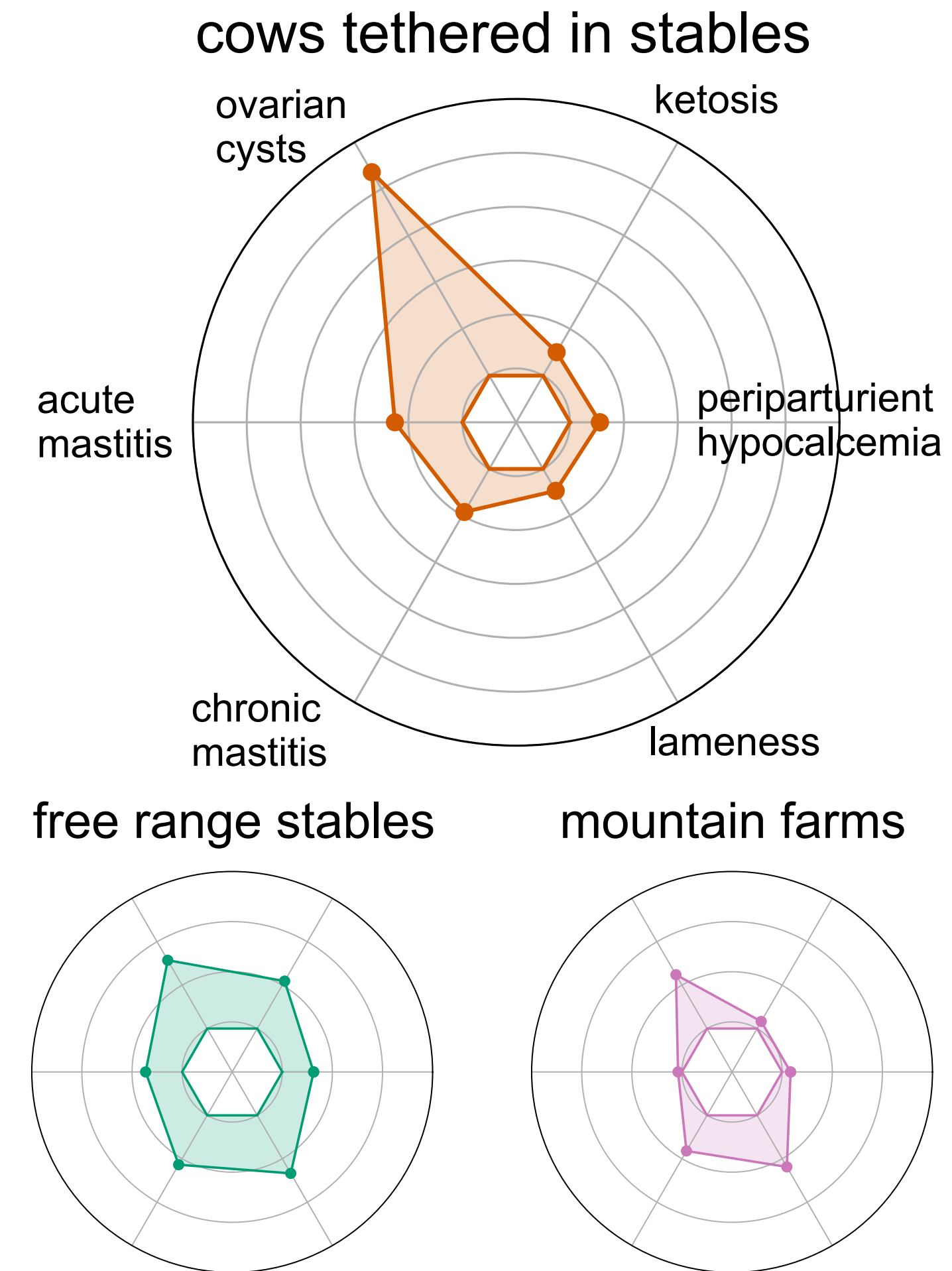
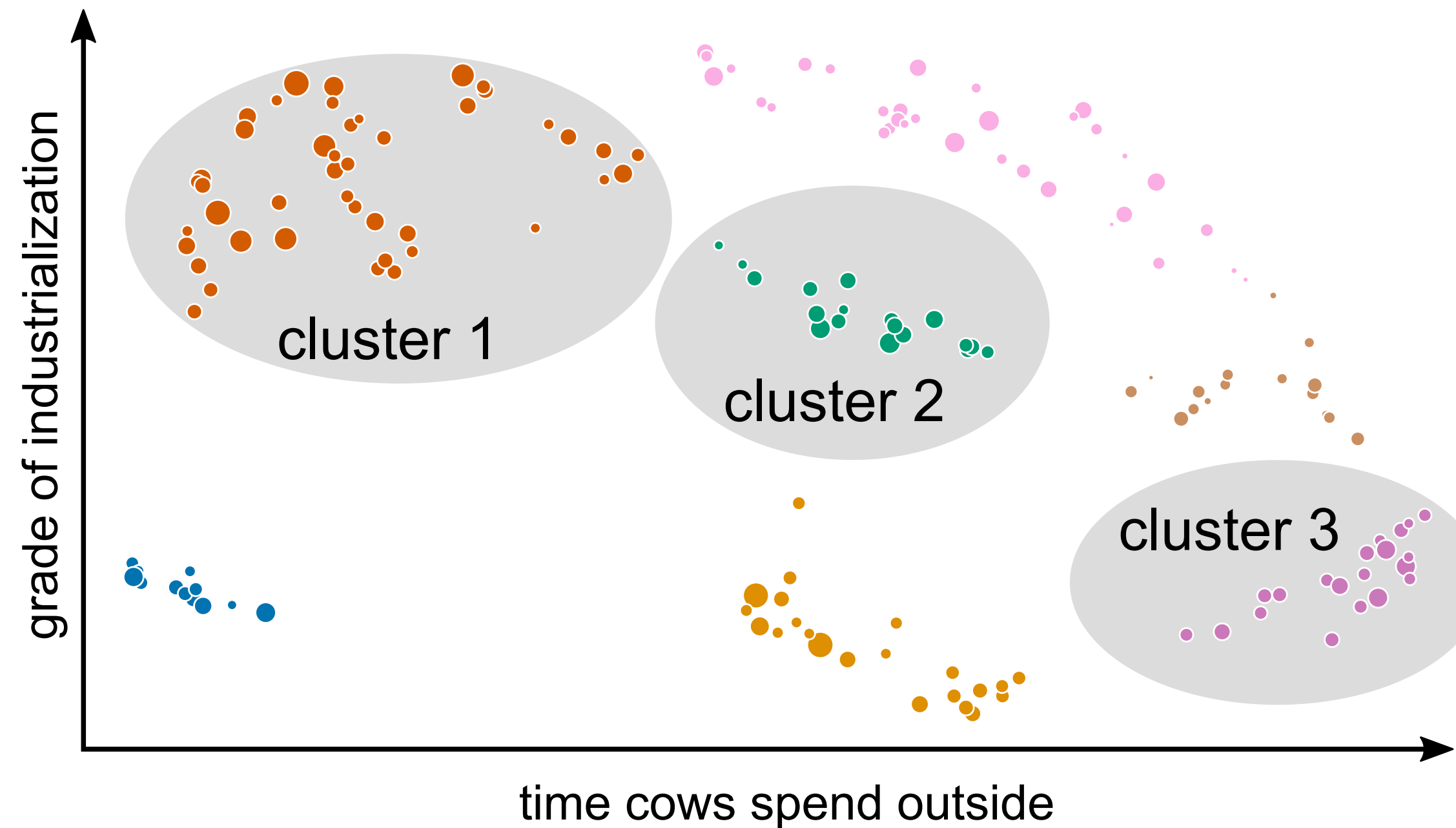
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Applications of Cluster Analysis: Summarizing Data

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What is Not Cluster Analysis?

- **Supervised Learning for Classification:** Groups or classes are known beforehand.
- **Grouping by a simple rule:** e.g., dividing students into groups alphabetically by their last names.

→ Cluster analysis uses only the attributes of observations to find a grouping.

Summary

- We apply cluster analysis or "unsupervised learning" when we have data **with attributes** but **without class membership** or target values.
- Cluster analysis is useful when we want to discover previously unknown patterns in data.
- Cluster analysis can also be helpful when we want to summarize the information contained in a data set.